

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AFISCIOEM05G - WASTE TO ENERGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Hazardous wastes have
 1. toxicity and reactivity
 2. ignitability and corrosivity
 3. both (1) and (2)
 4. none of the above
 - 2 Moisture content of MSW is determined by
 1. proximate analysis
 2. chemical analysis
 3. ultimate analysis
 4. COD and BOD
 - 3 Ethanol is used as
 1. solvent
 2. fuel
 3. catalyst
 4. both (1) and (2)
 - 4 The combustion of solid waste in incinerator is controlled by
 1. time, temperature and turbulence
 2. time, temperature and feed rate
 3. time, pressure and turbulence
 4. time, pressure and feed rate
 - 5 In which gasifier, moisture content of feed is not an important parameter?
 1. fluidized bed gasifier
 2. fixed bed gasifier
 3. plasma gasifier
 4. none of the above

- 6 Unit operation does not involve in harvesting algal biomass is
1. flocculation
 2. flotation
 3. sedimentation
 4. none of the above
- 7 What fuels can potentially be made from algae?
1. gasoline
 2. ethanol
 3. biodiesel and jet fuel
 4. all of the above
- 8 The _____ content in algae contributes to bio-oil
1. lipid
 2. water
 3. protein
 4. carbon dioxide
- 9 The detention time of an anaerobic digester operating at slow rate is
1. 5 to 10 days
 2. 10 to 20 days
 3. 30 to 60 days
 4. 5 to 6 months
- 10 Chemical detoxification can be carried out by using
1. sodium dithionite
 2. sodium sulfite
 3. sodium borohydride
 4. all of the above
- 11 Which separation technique is used to remove particles of size greater than 0.01 micro meter
1. gravity settler
 2. cyclone separator
 3. adsorption
 4. fabric filter
- 12 Overall efficiency of a power plant depends on
1. efficiency of boiler
 2. efficiencies of turbine and generator
 3. efficiency of works
 4. all of the above

Q2. Solve the following.

- A) What is waste? Give detailed classification of waste. 6
- B) Explain the factors affecting incineration. 6

Q3. Solve the following.

- A) Explain following types of gasifiers with neat diagrama. updraftb. Downdraft 6
- B) Explain Fischer Tropsch synthesis with the help of relevant chemical equations. 6
Give its applications and advantages.

- Q4. Solve Any Two of the following.
- A) Explain circulating fluidized bed pyrolytic reactor with neat diagram. 6
 - B) Discuss the properties of bio oil with their possible reason. Compare bio oil with diesel. 6
 - C) Explain different steps for production of ethanol from starchy crops 6
- Q5. Solve Any Two of the following.
- A) Explain construction and working of single stage and two stage anaerobic digesters. 6
 - B) Explain the factors affecting biogas yield. 6
 - C) State and explain different briquette characteristics. 6
- Q6. Solve Any Two of the following.
- A) Explain thermal and catalytic pyrolysis of plastic in detail. 6
 - B) Explain different parameters affecting MFC operation. 6
 - C) Write note on single step oil extraction from algal biomass 6

***** End *****